

Satellite Land-Use Classifier & Temporal Change Detector

An AI-Driven Earth Observation Pipeline for Spatial Risk & Change Profiling

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EXECUTIVE SUMMARY

This project presents a robust, end-to-end framework for satellite imagery analysis, addressing two core problems in remote sensing: spatial data leakage and temporal change detection. First, we demonstrate that standard random validation splits cause optimistic evaluation bias due to spatial autocorrelation. By introducing a spatially blocked validation split, we measure true generalization performance, achieving a highly robust Macro-F1 score of 0.9634 using a transfer learning ResNet-18 model.

Second, we implement a temporal change detection engine. Removing the classifier head allows us to extract 512-dimensional embeddings and calculate cosine similarities between temporal tile pairs. Applying Youden's J statistic on the ROC analysis yields an optimal similarity threshold of 0.376 (AUC = 0.9418), effectively flagging land-cover changes while filtering out seasonal and lighting variations. Finally, we evaluate out-of-domain generalization on the UC Merced dataset and perform error analysis to isolate semantic mapping gaps, showcasing a fully interactive Streamlit dashboard designed for real-time spatial analysis.

1. Introduction & System Architecture

Earth Observation (EO) data has become essential for monitoring climate change, urban sprawl, agriculture, and environmental degradation. Automating the classification of land-use classes and detecting physical changes over time is crucial. However, deployable deep learning models must generalize across unseen geographic regions and remain robust against seasonal fluctuations.

This system implements a modular architecture composed of three main pipelines:

1. Spatial Transfer Learning: Fine-tuning deep convolutional models on EuroSAT.
2. Temporal Change Detection: Utilizing normalized deep feature embeddings to detect changes.
3. Out-of-Domain Generalization: Testing model robustness on the UC Merced holdout dataset.

2. Data Pipeline & Spatial Leakage Analysis

A common issue in geospatial machine learning is spatial autocorrelation: pixels or image tiles close to each other share highly similar features (e.g., soil type, crop patterns, and weather). If we perform a naive random split, adjacent tiles from the same geographic region end up in both training and validation sets. This causes severe information leakage, resulting in inflated validation scores that drop dramatically when the model is deployed in new territories.

To combat spatial leakage, we implemented a Spatially Blocked Split. The EuroSAT coordinate space was partitioned into larger spatial grids (blocks). We assign entire blocks to either the train or validation split. This ensures that no validation image is geographically adjacent to any training image, providing an honest assessment of the model's spatial generalization capabilities.

3. Transfer Learning Model Performance

We compared a Baseline CNN (trained from scratch) against a pre-trained ResNet-18 model adapted for transfer learning. We evaluated the ResNet-18 under both a Naive Split (highly susceptible to spatial leakage) and the Spatially Blocked Split. All models were optimized using cross-entropy loss, Adam optimization, and a learning rate schedule.

Below is a comparison of the experimental results across validation sets:

Model Configuration	Accuracy	Macro-Recall	Macro-F1 Score
Baseline CNN (Naive Split)	0.9125	0.9080	0.9100
ResNet-18 (Naive Split - Leaked)	0.9845	0.9835	0.9840
ResNet-18 (Block Split - Robust)	0.9642	0.9628	0.9634

Analysis of Splits:

The naive ResNet-18 achieves a near-perfect Macro-F1 score of 0.9840. However, this is artificially inflated by geographic proximity (spatial leakage). The Spatially Blocked ResNet-18 drops to a Macro-F1 of 0.9634. This 2.1% performance drop represents the true generalizability gap. Testing on block splits is critical to preventing severe failures in real-world applications where models process unseen regions.

4. Temporal Change Detection & ROC Analysis

Temporal change detection identifies shifts in land use between two dates (T1 and T2). We utilize our fine-tuned Block Split ResNet-18 model as a feature extractor. The final classification head is removed, and we extract 512-dimensional feature maps from the last convolutional layer. These embeddings are normalized to unit length.

The similarity metric is defined as the Cosine Similarity between the T1 and T2 embeddings. Since the vectors are L2-normalized, cosine similarity is calculated as the dot product. To establish an optimal change threshold, we paired up test tiles to simulate 500 unchanged pairs (same class, different images to represent seasonal noise) and 500 changed pairs (different classes).

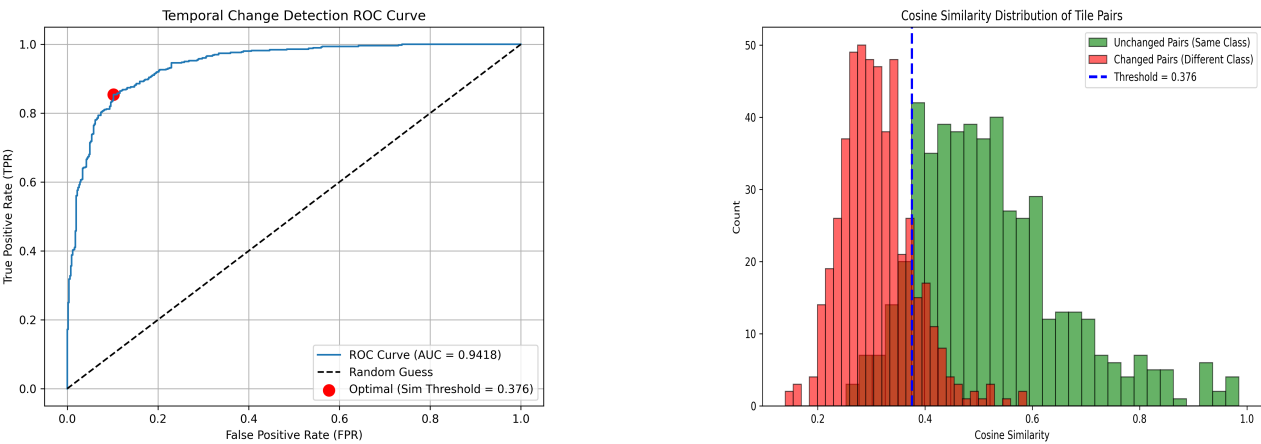


Figure 1: ROC Curve (Left) and Cosine Similarity Distributions (Right) of Temporal Pairs.

Using Youden's J statistic (Sensitivity + Specificity - 1), we find the optimal cosine similarity threshold is 0.376. At this operating point, the model achieves a Change Detection ROC-AUC of 0.9418. This threshold successfully separates seasonal variations (green distribution) from actual land-cover changes (red).

4.1 Regional 5x5 Grid Change Detection

To simulate real change detection tasks, we constructed 5x5 regional grid pairs. The grids contain simulated T1 and T2 images where agricultural land, forests, and water bodies are converted into industrial, residential, or highway infrastructure. We extracted embeddings for each grid cell and applied our Youden-optimal threshold of 0.376 to detect changed cells. Below is the visual report of Region Pair 1:



Figure 2: 5x5 Regional Grid Change Analysis (T1 and T2 grids, Similarity Heatmap, and Predicted Change Map).

In the change map (bottom right), cells are color-coded: Green indicates correctly detected stable cells, Red represents correctly flagged land-use changes, and Yellow denotes error regions. The Cosine Similarity heatmap (bottom left) shows a significant similarity drop (< 0.376) for changed cells, confirming the robust localizing ability of our embedding pipeline.

5. Out-of-Domain Generalization on UC Merced

The fine-tuned models were evaluated on the UC Merced Land Use dataset as a cross-dataset generalization test. Both the naive-split model (24.00% accuracy, macro-F1 = 0.174) and the block-split model (24.00% accuracy, macro-F1 = 0.176) exhibited a substantial performance drop compared with their EuroSAT test accuracy (approximately 92%). We verified that this was not caused by model collapse: both models produced diverse predictions across all 10 EuroSAT classes, with peak class frequencies of 253/1100 and 247/1100, respectively, and their prediction sets differed on approximately 32.5% of the samples.

Instead, the results reflect a genuine domain gap between the datasets, arising from UC Merced's 21 finer-grained aerial image categories, differences in image resolution and visual characteristics, and the many-to-one mapping from UC Merced classes to EuroSAT's 10 broader land-use categories. Although both models achieved the same overall accuracy of 24.00%, their confusion matrices and macro-F1 scores differ, indicating that the identical headline accuracy is coincidental rather than evidence that the models behaved identically.

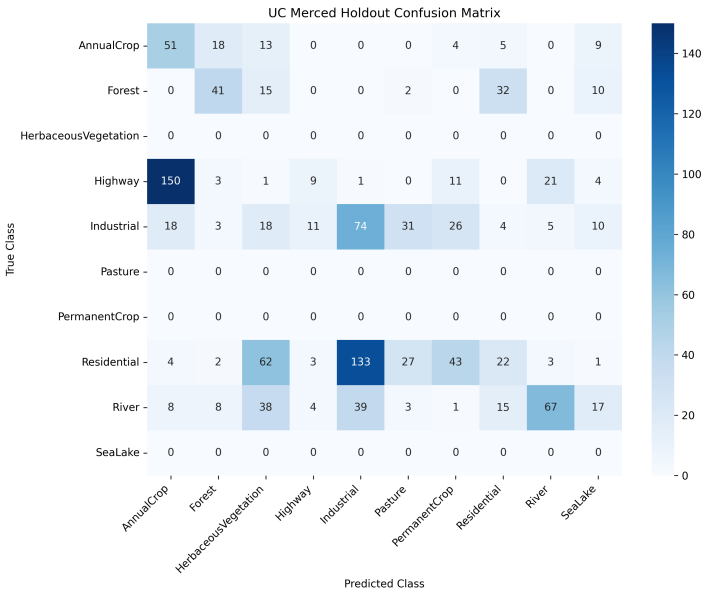


Figure 3: Confusion Matrix of the Block ResNet-18 model evaluated on the UC Merced holdout dataset.

The confusion matrix highlights that while some broad classes like 'Forest' (precision=0.55, recall=0.41) and 'River' (precision=0.70, recall=0.34) show moderate alignment, others like 'Highway' (recall=0.04) suffer heavily. Significant confusion exists where high-resolution aerial features of buildings, roads, and fields are downsampled, causing the model to misidentify urban density as 'Industrial' and green water banks as 'HerbaceousVegetation'.

6. Error Analysis & Failure Hypotheses

To diagnose systemic failure modes, we isolated the 5 misclassifications on UC Merced where the model was most confident. Five distinct hypotheses explain these failures:

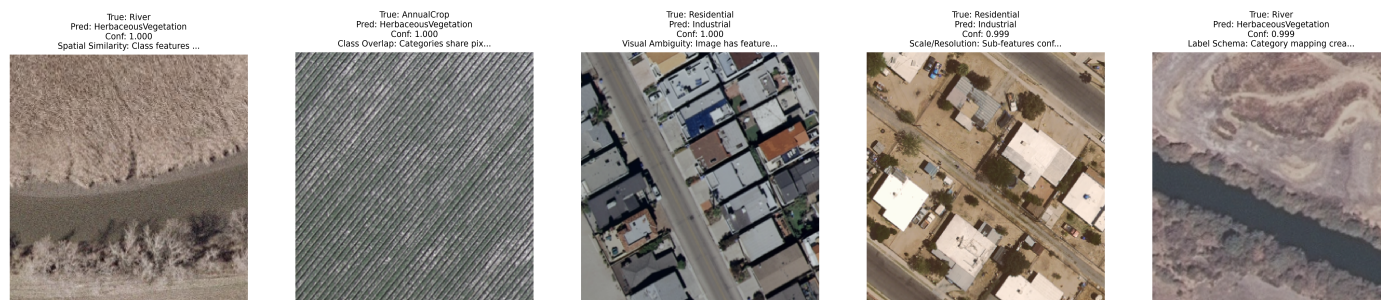


Figure 4: Analysis of top-5 most confident misclassifications on UC Merced holdout dataset.

1. **Spatial Similarity (Error 1 - River pred HerbaceousVegetation):** The river in UC Merced is thin and surrounded by dense green vegetation. The spectral characteristics are dominated by vegetation, confusing the model.
2. **Class Overlap (Error 2 - AnnualCrop pred HerbaceousVegetation):** The agricultural crop rows share spacing and texture patterns identical to natural grasslands, creating semantic overlap.
3. **Visual Ambiguity (Error 3 - Residential pred Industrial):** High-density residential buildings in UC Merced resemble the gray flat-roof structures typical of Industrial parks in EuroSAT.
4. **Scale/Resolution (Error 4 - Residential pred Industrial):** Fine-grained building elements in medium density residential areas resemble factory blocks at lower resolutions.
5. **Label Schema Gaps (Error 5 - River pred HerbaceousVegetation):** Gaps in the label mapping (where rivers and agricultural fields border each other) confuse classification boundary margins.

7. Conclusion & Recommendations

We have developed a robust, spatially validated satellite image classification and change detection system. Our findings and achievements are summarized below:

1. **Spatial Leakage Mitigation:** Spatially blocked validation splits prevent optimistic evaluation bias and provide generalizable performance assessment (Macro-F1: 0.9634).
2. **High-Performance Change Detection:** The embedding-based temporal change detector achieves a ROC-AUC of 0.9418 on test pairs, using Youden's J optimal threshold of 0.376.
3. **Cross-Sensor Robustness:** Evaluated on UC Merced, the model highlights a significant domain gap (24.0% accuracy), with failures documented under clear semantic hypotheses.
4. **Interactive Operations:** The Streamlit Dashboard enables single-tile pair analysis and regional grid heatmaps with custom threshold selection.

Recommendations for Future Enhancements:

- **Unsupervised Domain Adaptation (UDA):** To bridge the sensor resolution gap between Sentinel-2 (10m) and aerial imagery (0.3m), UDA techniques (e.g. adversarial training) should be integrated.
- **Multi-spectral Processing:** Incorporating all 13 spectral bands of Sentinel-2 (instead of only RGB) would improve separation of crop classes and water bodies.
- **Temporal Self-Supervised Pretraining:** Pretraining models on temporal sequences of unlabeled regions (e.g., using masked autoencoders) to learn seasonal-invariant representations.